

$$1 \quad \frac{dz}{dt} = e^{-t} \sin(t)(2 \cos t - \sin t)$$

$$3 \quad 2 \cos \left(\frac{2t}{1-t^2} \right) \frac{1+t^2}{(1-t^2)^2}$$

$$5 \quad 2e^{1-t^2}(1-2t^2)$$

$$7 \quad \frac{\partial z}{\partial u} = \frac{1}{vu} \cos \left(\frac{\ln u}{v} \right) \\ \frac{\partial z}{\partial v} = -\frac{\ln u}{v^2} \cos \left(\frac{\ln u}{v} \right)$$

$$9 \quad \frac{\partial z}{\partial u} = \frac{e^v}{u} \\ \frac{\partial z}{\partial v} = e^v \ln u$$

$$11 \quad \frac{\partial z}{\partial u} = 2ue^{(u^2-v^2)}(1+u^2+v^2) \\ \frac{\partial z}{\partial v} = 2ve^{(u^2-v^2)}(1-u^2-v^2)$$

$$13 \quad \frac{\partial z}{\partial u} = (e^{-v} \cos u - v(\cos u)e^{-u \sin v}) \sin v \\ - (-u(\sin v)e^{-v \cos u} + e^{-u \sin v})v \sin u \\ \frac{\partial z}{\partial v} = (e^{-v} \cos u - v(\cos u)e^{-u \sin v})u \cos v \\ + (-u(\sin v)e^{-v \cos u} + e^{-u \sin v}) \cos u$$

$$15 \quad \frac{\partial z}{\partial u} = \frac{-2uv^2}{u^4 + v^4} \\ \frac{\partial z}{\partial v} = \frac{2vu^2}{u^4 + v^4}$$

$$17 \quad -2\rho \cos 2\phi, 0$$

$$19 \quad 401.1$$

$$21 \quad -5 \text{ pascal/hour}$$

$$23 \quad -0.6$$

$$25 \quad \text{Three}$$

$$27 \quad \frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

$$29 \quad b \cdot e + d \cdot p$$

$$35 \quad b \cdot e + d \cdot p$$

$$37 \quad (a) \quad \frac{\partial z}{\partial r} = \cos \theta \frac{\partial z}{\partial x} + \sin \theta \frac{\partial z}{\partial y} \\ \frac{\partial z}{\partial \theta} = r(\cos \theta \frac{\partial z}{\partial y} - \sin \theta \frac{\partial z}{\partial x}) \\ (b) \quad \frac{\partial z}{\partial y} = \sin \theta \frac{\partial z}{\partial r} + \frac{\cos \theta}{r} \frac{\partial z}{\partial \theta} \\ \frac{\partial z}{\partial x} = \cos \theta \frac{\partial z}{\partial r} - \frac{\sin \theta}{r} \frac{\partial z}{\partial \theta}$$

$$39 \quad \left(\frac{\partial U_3}{\partial P} \right)_V$$

$$41 \quad \left(\frac{\partial U}{\partial T} \right)_V = 7/2 \\ \left(\frac{\partial U}{\partial V} \right)_T = 11/4$$

$$45 \quad \int_0^b F_u(x, y) dy$$

$$49 \quad C_R \text{ and } C_T \text{ must be evaluated at } (5, 1)$$

$$51 \quad g(t) = 3 + t, h(t) = t$$

$$53 \quad z = x + y, x = e^t \text{ and } y = t^2, \text{ many other answers are possible}$$

$$55 \quad z = x + y, x = t \text{ and } y = t$$

$$5 \quad f_{xx} = 0 \\ f_{yx} = e^y = f_{xy} \\ f_{yy} = e^y(x+2+y)$$

$$7 \quad f_{xx} = -\left(\sin \left(\frac{x}{y}\right)\right) \left(\frac{1}{y^2}\right) \\ f_{xy} = -\left(\sin \left(\frac{x}{y}\right)\right) \left(\frac{-x}{y^2}\right) \left(\frac{1}{y}\right) \\ + \left(\cos \left(\frac{x}{y}\right)\right) \left(\frac{-1}{y^2}\right) = f_{yx} \\ f_{yy} = -\left(\sin \left(\frac{x}{y}\right)\right) \left(\frac{-x}{y^2}\right)^2 \\ + \left(\cos \left(\frac{x}{y}\right)\right) \left(\frac{2x}{y^3}\right)$$

$$9 \quad f_{xx} = 30xy^2 + 18 \\ f_{xy} = 30x^2y - 21y^2 \\ f_{yx} = 30x^2y - 21y^2 \\ f_{yy} = 10x^3 - 42xy$$

$$11 \quad f_{xx} = -12 \sin 2x \cos 5y \\ f_{xy} = -30 \cos 2x \sin 5y \\ f_{yx} = -30 \cos 2x \sin 5y \\ f_{yy} = -75 \sin 2x \cos 5y$$

$$13 \quad Q(x, y) = 1 + 2x - 2y + x^2 - 2xy + y^2$$

$$15 \quad Q(x, y) = 1 + x + x^2/2 - y^2/2$$

$$17 \quad Q(x, y) = 1 - x^2/2 - 3xy - (9/2)y^2$$

$$19 \quad Q(x, y) = -y + x^2 - y^2/2$$

$$21 \quad 1 + x - y/2 - x^2/2 + xy/2 - y^2/8$$

$$23 \quad (a) \quad \text{Negative}$$

$$(b) \quad \text{Zero}$$

$$(c) \quad \text{Negative}$$

$$(d) \quad \text{Zero}$$

$$(e) \quad \text{Zero}$$

$$25 \quad (a) \quad \text{Positive}$$

$$(b) \quad \text{Zero}$$

$$(c) \quad \text{Positive}$$

$$(d) \quad \text{Zero}$$

$$(e) \quad \text{Zero}$$

$$27 \quad (a) \quad \text{Zero}$$

$$(b) \quad \text{Negative}$$

$$(c) \quad \text{Zero}$$

$$(d) \quad \text{Negative}$$

$$(e) \quad \text{Zero}$$

$$29 \quad (a) \quad \text{Positive}$$

$$(b) \quad \text{Positive}$$

$$(c) \quad \text{Zero}$$

$$(d) \quad \text{Zero}$$

$$(e) \quad \text{Zero}$$

$$31 \quad (a) \quad \text{Positive}$$

$$(b) \quad \text{Negative}$$

$$(c) \quad \text{Negative}$$

$$(d) \quad \text{Negative}$$

$$(e) \quad \text{Positive}$$

$$33 \quad L(x, y) = y$$

$$Q(x, y) = y + 2(x-1)y$$

$$L(0.9, 0.2) = 0.2$$

$$Q(0.9, 0.2) = 0.16$$

$$f(0.9, 0.2) = 0.162$$

$$39 \quad a = -b^2$$

$$41 \quad \text{Positive, negative}$$

$$43 \quad (a) \quad z_{yx} = 4y$$

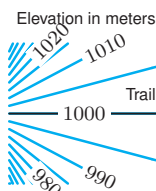
$$(b) \quad z_{xyx} = 0$$

$$(c) \quad z_{xyy} = 4$$

$$45 \quad d = e = f = 0$$

$$47 \quad d = 0, e > 0, f < 0$$

$$49 \quad (a)$$



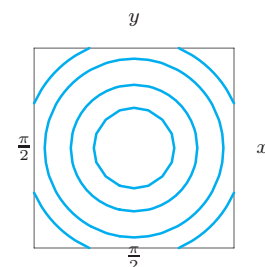
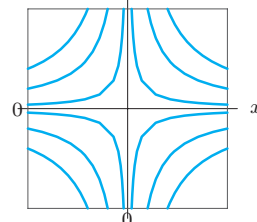
$$(b) \quad \frac{\partial h}{\partial x} = 0, \quad \frac{\partial h}{\partial y} > 0, \\ (\partial^2 h)/(\partial x \partial y) < 0$$

$$(c) \quad (\partial^2 h)/(\partial x \partial y)$$

$$51 \quad (a) \quad A$$

$$(b) \quad B$$

$$53 \quad (a) \quad xy \\ 1 - \frac{1}{2}(x - \frac{\pi}{2})^2 - \frac{1}{2}(y - \frac{\pi}{2})^2 \\ (b)$$



$$55 \quad (a) \quad (i) \quad \text{Dollars/Year, negative,} \quad (ii) \quad \text{Dollars/Dollar, positive}$$

$$(b) \quad \partial^2 P/(\partial A \partial C) < 0$$

$$(c) \quad eCA$$

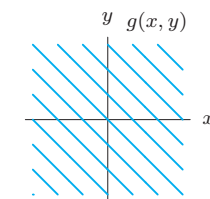
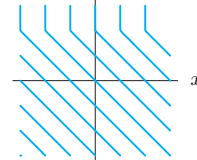
$$57 \quad (a) \quad 1 + x/2 + y$$

$$(b) \quad 1 + x/2 + y - x^2/8 - xy/2 - y^2/2$$

$$61 \quad \text{Counterexample: } f(x, y) = x^3 + y^4$$

$$63 \quad f(x, y) = x^2 + y^2$$

$$65 \quad y \quad f(x, y)$$



Section 14.7

$$1 \quad f_{xx} = 2$$

$$f_{yy} = 2$$

$$f_{yx} = 2$$

$$f_{xy} = 2$$

$$3 \quad f_{xx} = 6y$$

$$f_{xy} = 6x + 15y^2$$

$$f_{yx} = 6x + 15y^2$$

$$f_{yy} = 30xy$$

Section 14.8

$$1 \quad (0, 0)$$

$$3 \quad x\text{-axis and } y\text{-axis}$$

$$5 \quad \text{None}$$